Final

Sarah Whitney

2025-05-20

```
N       <- 2000      # pop. size
var_X   <- 0.40      # additive variance
var_N   <- 0.40      # nutrition variance
var_E   <- 0.20      # residual environmental variance
s       <- 0.30      # strength of the fitness effect of nutrition
n_gen   <- 20        # generations to follow (for the "no response" panel)
h2      <- var_X / (var_X + var_N + var_E)   # realised heritability
h2
```

```
## [1] 0.4
```

```
## one generation, returns a data.frame of individuals
draw_generation <- function(N, var_X, var_N, var_E) {
  tibble(
    X = rnorm(N, 0, sqrt(var_X)),
    N = rnorm(N, 0, sqrt(var_N)),
    E = rnorm(N, 0, sqrt(var_E))
  ) %>%
    mutate(
      P = X - N + E,
      w = 1 + s * N       # linear, always positive if s < ~3/SD(N)
    )
}

## mid-parent regression slope approx h²
midparent_offspring <- function(df){
  # random pairing
  moms   <- sample(nrow(df))
  dads   <- sample(nrow(df))
  midP   <- (df$P[moms] + df$P[dads]) / 2
  midX   <- (df$X[moms] + df$X[dads]) / 2

  # offspring inherit only X; nutrition & E are fresh draws
  X_off  <- rnorm(length(midX), mean = midX, sd = sqrt(var_X/2))
  N_off  <- rnorm(length(midX), 0, sqrt(var_N))
  E_off  <- rnorm(length(midX), 0, sqrt(var_E))
  P_off  <- X_off - N_off + E_off

  tibble(midP, P_off)
}
```

```r
gen0  <- draw_generation(N, var_X, var_N, var_E)
regdf <- midparent_offspring(gen0)
slp   <- coef(lm(P_off ~ midP, data = regdf))[2]

pA <- ggplot(regdf, aes(midP, P_off)) +
  geom_point(alpha = 0.3, colour = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, colour = "darkblue") +
  labs(x = "Mid-parent phenotype",
       y = "Offspring phenotype",
       title = "(A) Heritability") +
  annotate("text", x = Inf, y = -Inf, hjust = 1.05, vjust = -0.6,
           label = sprintf("slope approx h² = %.2f", slp)) +
  theme_minimal(base_size = 11)
```

```r
cov_pw <- cov(gen0$P, gen0$w)

pB <- ggplot(gen0, aes(P, w)) +
  geom_point(alpha = 0.3, colour = "tomato") +
  geom_smooth(method = "lm", se = FALSE, colour = "firebrick") +
  labs(x = "Phenotype",
       y = "Fitness",
       title = "(B) Covariance between P and w") +
  annotate("text", x = Inf, y = -Inf, hjust = 1.05, vjust = -0.6,
           label = sprintf("Cov(P , w) = %.3f", cov_pw)) +
  theme_minimal(base_size = 11)
```

```r
mean_P <- numeric(n_gen + 1)
gen     <- gen0

mean_P[1] <- mean(gen$P)

for (g in 1:n_gen){
  ##  selection proportional to w
  sel_idx <- sample(nrow(gen), N, replace = TRUE, prob = gen$w)
  parents <- gen[sel_idx, ]

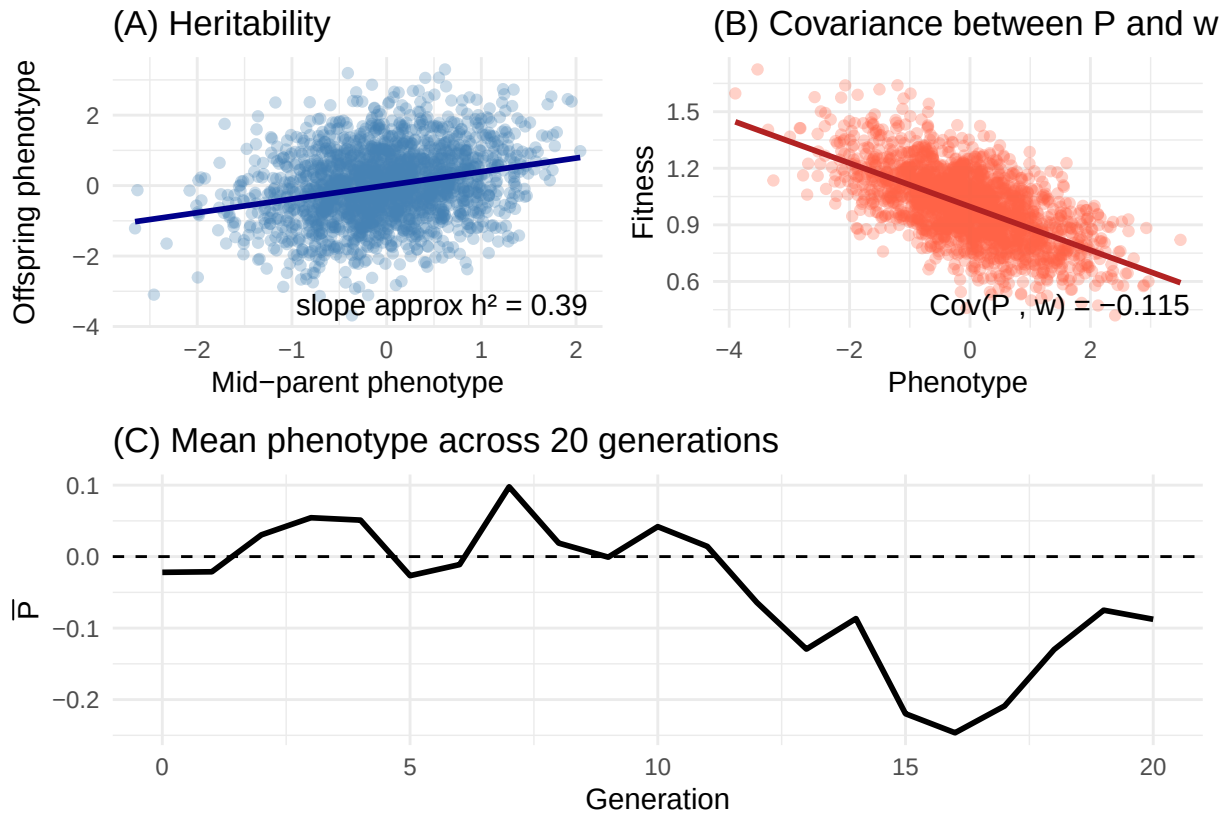
  ##  offspring genetics come only from X (nutrition/env. are new each time)
  X_off <- rnorm(N, parents$X, sd = sqrt(var_X/2))
  N_off <- rnorm(N, 0, sqrt(var_N))
  E_off <- rnorm(N, 0, sqrt(var_E))

  gen <- tibble(X = X_off, N = N_off, E = E_off,
                P = X_off - N_off + E_off,
                w = 1 + s * N_off)
  mean_P[g + 1] <- mean(gen$P)
}

pC <- tibble(Generation = 0:n_gen, MeanP = mean_P) %>%
  ggplot(aes(Generation, MeanP)) +
  geom_line(colour = "black", linewidth = 1) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(title = "(C) Mean phenotype across 20 generations",
       x = "Generation", y = expression(bar(P))) +
```

```
  theme_minimal(base_size = 11)
```

```
(pA | pB) / pC
```




```
## Parameters for stabilizing selection
omega  <- 0.5    # strength of stabilizing selection
N_opt  <- 0.5    # optimal nutrition value

stab_w <- function(N){
  exp(-((N - N_opt)^2) / (2 * omega^2))
}

draw_generation_stab <- function(N, var_X, var_N, var_E) {
  tibble(
    X = rnorm(N, 0, sqrt(var_X)),
    N = rnorm(N, 0, sqrt(var_N)),
    E = rnorm(N, 0, sqrt(var_E))
  ) %>%
  mutate(
    P = X - N + E,
    w = stab_w(N)
  )
}
```

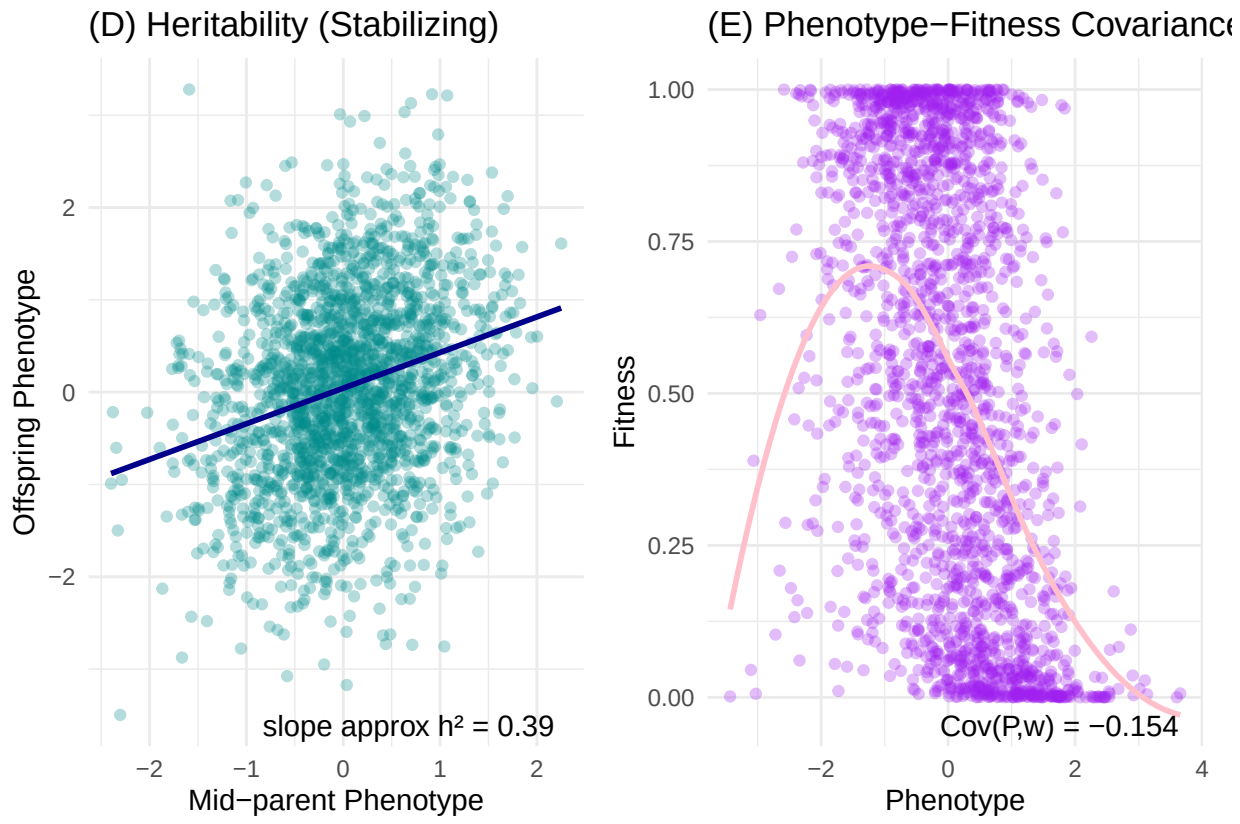
```r
gen0_stab <- draw_generation_stab(N, var_X, var_N, var_E)
regdf_stab <- midparent_offspring(gen0_stab)

slp_stab <- coef(lm(P_off ~ midP, data = regdf_stab))[2]
cov_pw_stab <- cov(gen0_stab$P, gen0_stab$w)

pA_stab <- ggplot(regdf_stab, aes(midP, P_off)) +
  geom_point(alpha = 0.3, colour = "darkcyan") +
  geom_smooth(method = "lm", se = FALSE, colour = "darkblue") +
  labs(x = "Mid-parent Phenotype", y = "Offspring Phenotype",
       title = "(D) Heritability (Stabilizing)") +
  annotate("text", x = Inf, y = -Inf, hjust = 1.1, vjust = -0.5,
           label = sprintf("slope approx h² = %.2f", slp_stab)) +
  theme_minimal()

pB_stab <- ggplot(gen0_stab, aes(P, w)) +
  geom_point(alpha = 0.3, colour = "purple") +
  geom_smooth(method = "loess", se = FALSE, colour = "pink") +
  labs(x = "Phenotype", y = "Fitness",
       title = "(E) Phenotype-Fitness Covariance (Stabilizing)") +
  annotate("text", x = Inf, y = -Inf, hjust = 1.1, vjust = -0.5,
           label = sprintf("Cov(P,w) = %.3f", cov_pw_stab)) +
  theme_minimal()

pA_stab | pB_stab
```



```r
mean_P_stab <- numeric(n_gen + 1)
gen_stab <- gen0_stab
mean_P_stab[1] <- mean(gen_stab$P)

for (g in 1:n_gen){
  sel_idx <- sample(nrow(gen_stab), N, replace = TRUE, prob = gen_stab$w)
  parents <- gen_stab[sel_idx,]

  X_off <- rnorm(N, parents$X, sd = sqrt(var_X/2))
  N_off <- rnorm(N, 0, sqrt(var_N))
  E_off <- rnorm(N, 0, sqrt(var_E))

  gen_stab <- tibble(X = X_off, N = N_off, E = E_off,
                     P = X_off - N_off + E_off,
                     w = stab_w(N_off))

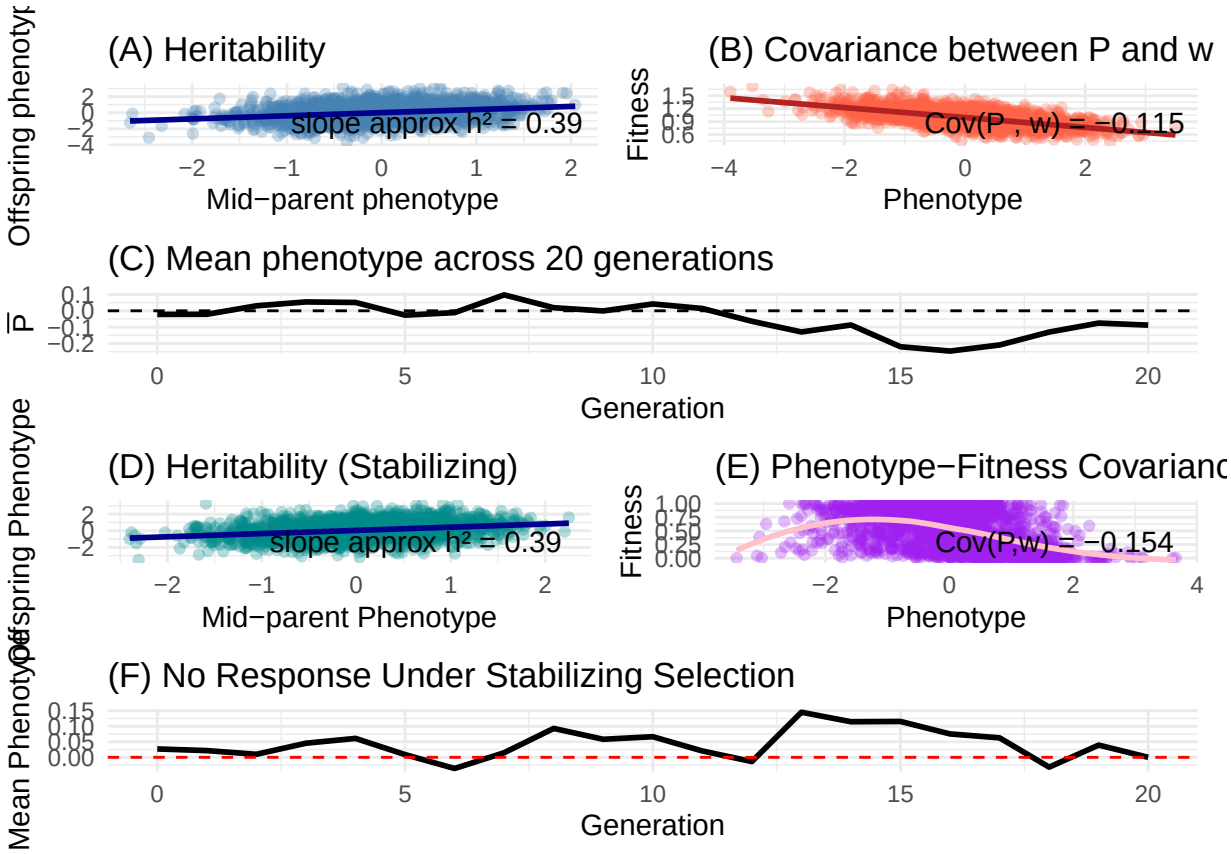
  mean_P_stab[g+1] <- mean(gen_stab$P)
}

pC_stab <- tibble(Generation = 0:n_gen, MeanP = mean_P_stab) %>%
  ggplot(aes(Generation, MeanP)) +
  geom_line(colour = "black", linewidth = 1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "(F) No Response Under Stabilizing Selection",
       x = "Generation", y = "Mean Phenotype") +
  theme_minimal()

pC_stab
```




```
(pA | pB) / pC /
(pA_stab | pB_stab) / pC_stab
```



The results of the above simulations of linear and stabilizing selection demonstrate scenarios in which heritability, covariance between phenotype and fitness, and response to selection interact in complex ways. Under linear selection, these simulations exhibit substantial heritability (h^2 is approximately 0.4), which is consistent with the realized heritability calculated from the provided variance components (additive genetic variance, nutrition, and residual environmental variance). The regression plot (Fig. A) demonstrates that offspring phenotypes strongly correlate with mid-parent phenotypes. Additionally, the covariance between phenotype and fitness (Fig. B) is negative, indicating an inverse relationship between the selected phenotype and fitness. This is the scenario deliberately chosen to "break the model." This shows how covariance influences evolutionary trajectories and responses to selection. Additionally, in Figure C, linear selection model shows no response over generations despite the presence of both heritability and phenotype-fitness covariance. This highlights the idea that simply the presence of heritability and covariance with fitness does not guarantee directional evolutionary change if selection acts indirectly (due to non-genetic or confounding environmental factors). This result emphasizes the importance of distinguishing between direct genetic selection and correlated responses. With the stabilizing selection scenario, by modeling fitness as a Gaussian function centered on an optimal phenotype (optimal nutrition), the simulation captures stabilizing selection's maintaining trait variance around an optimum. Here, despite a similarly high heritability (Fig. D), there is still no directional response across generations (Fig. F). This supports the distinction between stabilizing and directional selection. Overall, these simulations illustrate the following: heritability is a necessary but insufficient condition for evolutionary change and the evolutionary implications of linear vs stabilizing selection.